

FENGLING YANG

✉ fenglingyang@stu.pku.edu.cn · ☎ (+86) 158-0868-7192 · Peking University

EDUCATION

Peking University, B.S. Candidate in Computer Science and Technology 2023 - Present

Honors and Awards

- Peking University Freshman Scholarship
- Dean's Freshman Scholarship, School of Electronics Engineering and Computer Science, Peking University
- Peking University Merit Student
- Peking University Lingjun Leadership Scholarship
- Ranked 6/43 in the comprehensive evaluation for the 2024 - 2025 academic year

RESEARCH EXPERIENCE

Peking University YC Lab 2026.03 - present

- Conducted research on Genome Foundation Models and AI for Science.
- Co-authored a Perspective paper proposing a closed-loop driven DNA design system framework, involving literature review, manuscript writing, and figure design. **Planned submission to The Innovation (Cell Press).**

Peking University VCL Lab (Visual Computing & Learning) 2025.10 - 2026.03

- Participated in a 3D scene reconstruction project, responsible for developing core modules of the rendering pipeline. Implemented automated scripts using the Blender Python API, enabling end-to-end automation from 3D data import, scene configuration (lighting, cameras, materials), to large-scale batch rendering.
- **Submission to SIGGRAPH Asia 2026.**

Peking University VIE Group 2025.04 - 2025.10

- Worked on surgical video understanding, focusing on instrument segmentation and visual perception in complex surgical scenarios.
- Improved state-of-the-art semantic segmentation models using public benchmark datasets (EndoVis17/18) and real clinical data, enhancing instrument recognition and segmentation performance.

PROJECT EXPERIENCE

XV6 Operating System Kernel Labs Sep. 2025 - Dec. 2025

- Implemented core operating-system functionalities in XV6, including system calls, process scheduling, and memory management.

Computer Networks Course Projects Sep. 2025 - Dec. 2025

- **Application-Layer Protocol Design (myFTP):** Designed and implemented a custom file transfer protocol using POSIX APIs and TCP sockets. Built a client-server architecture with multi-process concurrency and used fork/execv to invoke system utilities such as ls and sha256sum. Supported file upload and download, directory navigation, and SHA-256 integrity verification.
- **Reliable Transport Protocol (RTP):** Designed and implemented a TCP-like reliable transport protocol over unreliable UDP, including connection establishment and termination, sequence-number management, timeout detection, and retransmission. Handled packet loss, bit corruption, packet reordering, and duplication in simulated network channels.

- **Network Simulation and Analysis (ns-3):** Built network simulation environments using ns-3 and developed scripts to generate traffic under different network topologies. Analyzed throughput and latency from simulation logs and evaluated protocol performance under different congestion scenarios.

Introduction to Computer Systems (CSAPP) Labs

Sep. 2024 - Dec. 2024

- Completed the course laboratories in C on a Linux virtual machine.
- **Malloc Lab:** Implemented a dynamic memory allocator.
- **Cache Lab:** Developed a cache simulator and optimized matrix-transpose operations to minimize simulated cache misses.
- **Proxy Lab:** Implemented a concurrent, multithreaded Web proxy with object caching.

ACTIVITIES

- Participated in the Computer Science “Growth Partners” Summer School organized by the Institute for Interdisciplinary Information Sciences(Yao Class), Tsinghua University, from June 23 to July 4, 2025.
- Participated in the social practice program organized by the School of Electronics Engineering and Computer Science, Peking University, from August 25 to August 28, 2024.